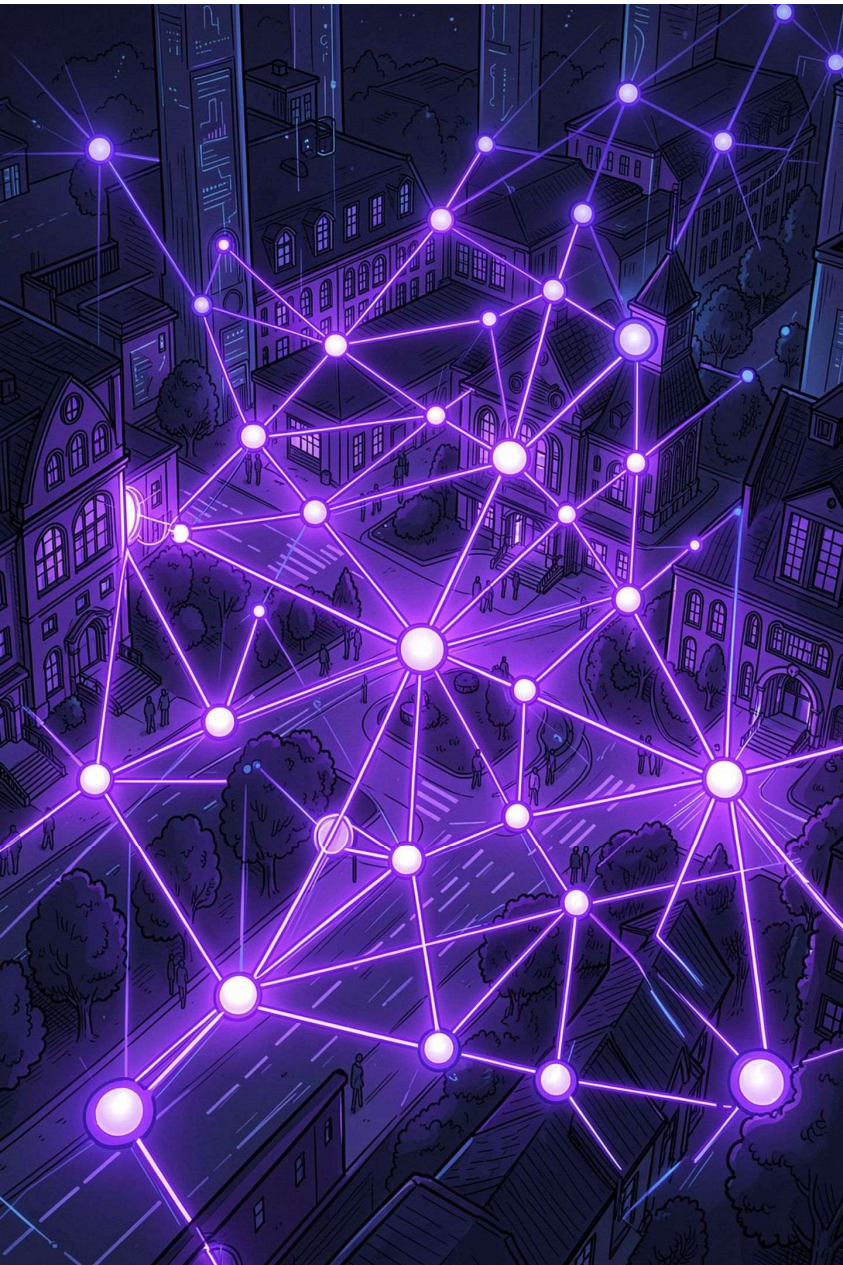




UCS503 Software Engineering Project

CampusConnect Verified Academic and Social Networking System

A role-aware platform that connects campus communication, course materials, and AI content moderation in one place.

The Problem with Fragmented Campus Communication

Informal tools like WhatsApp fall short on security, organization, and accountability in educational institutions.

No Identity Verification

Without domain verification, outsiders can join groups, impersonate faculty, and share leaked papers with no accountability.

Information Overload

Course material, lab manuals, and updates get buried in chat threads with no structure or search capability.

No Role-Based Access Control

Teachers end up sharing personal contacts, and students cannot tell official announcements apart from casual messages.

No Governance or Moderation

Institutions have no way to detect or act on harassment, hate speech, or academic dishonesty automatically.



Architecture

Full-Stack System Design and Infrastructure

Designed for type safety, relational integrity, real-time updates, and AI-assisted content safety.

Frontend and UI Layer

Next.js 16 App Router , React 19 Server Components ,
TypeScript & Tailwind CSS , Radix UI primitives

Data Layer

PostgreSQL on Neon Cloud , Prisma 7 ORM , 14-table
relational schema , 7 domain enums , cascade
constraints

Media and Real-Time

Cloudinary CDN API , Socket.io WebSocket gateway ,
instant notifications & live peer feeds

AI Safety Engine

Google Gemini 2.5 Flash , post/comment ingestion
pipelines , multi-tier sensitivity classification

Features

Key Features and Capabilities

Four core modules creating a secure and interactive campus experience.



Verified Auth and Token-Based Invites

NextAuth.js with bcrypt hashing. Admin invite tokens are signed, capped by usage, and auto-assign user roles.



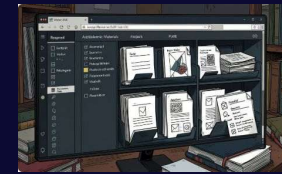
Social Feed and Friend Network

Paginated feed with threaded discussions, likes, and content reports. Includes bidirectional friend requests and peer suggestions.



Academic Groups and Collaboration

Teachers create groups for subjects, batches, labs, and clubs with member directories and granular role management.



Course Materials and PDF Preview

Materials organized into folders: Lectures, Assignments, Lab Notes, Syllabus. Built-in PDF viewer, no download needed.

What We Built and What Comes Next

Implemented and Verified (Phases 0-5)

- **Phases 0-1:** Full Prisma schema (14 models), Neon PostgreSQL setup, DB seeding
- **Phase 2:** Prisma, NextAuth, Cloudinary, Gemini AI singletons and route middleware
- **Phase 3:** Login, registration, email verification, token-based invite system
- **Phase 4:** Dashboard, profiles, friend network, community feed, Gemini content flagging
- **Phase 5:** Academic groups, member directory, materials hub, PDF previewer

✓ Full test suite passing. Zero TypeScript errors. Production build verified.

Up Next (Phases 6-10)

- **Phase 6:** Interactive academic calendar with color-coded lecture, lab, and exam filters
- **Phase 7:** Real-time 1-on-1 chat with live typing indicators via WebSocket
- **Phase 8:** Admin dashboard for user management, join policies, and moderation
- **Phase 9:** Dynamic Gemini sensitivity tuning and violation reporting
- **Phase 10:** Vercel Edge deployment, DB indexing, and GitHub Actions CI/CD

Impact

Value Delivered to Each Stakeholder

Students

One place to access course material, connect with batchmates, and never miss a deadline.

Faculty and Educators

A dedicated space to share course material and communicate without mixing work with personal contact details.

Institution Administration

Complete access control, automated content moderation, and reduced manual administrative work.

100%

Type-Safe Codebase

Strict TypeScript, route guards, and sub-second page loads

14

DB Models

Relational schema with cascade constraints

0

TS Errors

Production build fully validated

Ready for Live Demo and Code Walkthrough

Repository: github.com/aayush-10k/ucs503p-202627odd-CampusConnect